

# Amaan Pathan

Backend Engineer | Golang · TypeScript · Node.js · PostgreSQL · Redis · Docker · Distributed Systems

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## SUMMARY

Backend Engineer with open source experience at BookBrainz and Guidefox and production freelance work at NIAT serving 500+ active users. Proficient in Golang, TypeScript/Node.js, PostgreSQL, Redis, and large-scale backend architecture. Delivered 3+ end-to-end backend systems including a multi-step workflow automation engine and a fully functional HTTP/1.1 server built from raw TCP sockets. Committed to clean architecture, API security, and measurable performance improvements.

## SKILLS

- **Languages:** Golang, TypeScript, JavaScript (ES6+), Python, SQL
- **Backend:** RESTful APIs, Distributed Systems, System Design, JWT Authentication, RBAC, NATS JetStream, Workflow Automation, Microservices, Node.js, Express.js
- **Databases:** PostgreSQL, Redis, MongoDB; schema design, query optimization, connection pooling, Prisma ORM
- **Infrastructure & DevOps:** Docker, Linux/Unix, Bash, Nginx, Kubernetes, AWS, Cloudflare, CI/CD pipelines, Git, GitHub
- **Networking & Systems:** HTTP/1.1, TCP/IP, OS internals, concurrency, goroutines, connection lifecycle management
- **Tools & Practices:** Postman, VS Code, Zod, DTO-based API contracts, unit & integration testing

## WORK EXPERIENCE

### Open Source Contributor | BookBrainz | Remote

Jan. 2026 – Mar. 2026

- Replaced external Docker wait-scripts with Docker-native health checks across 3 services (PostgreSQL, Elasticsearch, Redis), eliminating startup race conditions and reducing environment setup failures by ~90%.
- Resolved GitHub Actions CI failures caused by Docker version incompatibilities on `ubuntu-latest` runners, restoring a 100% CI pass rate and unblocking the project's active PR review pipeline.

### Software Engineer | NIAT (Freelance / Contract) | Remote

Sep. 2025 – Oct. 2025

- Designed RESTful backend services for an internal platform with 500+ active users; delivered authentication, RBAC, and a RAG pipeline improving query accuracy by 35%.
- Optimized API response times to under 120ms under concurrent load through query tuning and targeted caching strategies.

### Open Source Contributor | Guidefox | Remote

Oct. 2024 – Jan. 2025

- Integrated Zod-based validation across 20+ backend APIs, reducing runtime errors by 40%; introduced DTO-based contracts cutting duplicated backend logic by 30%.
- Expanded test coverage from 55% to 80% through targeted unit and integration tests, cutting defect escape rate and accelerating the project's CI merge cycles.

## PROJECTS

### Hermes – Fault-Tolerant Workflow Automation Engine [GitHub](#) Golang · PostgreSQL · Redis · Docker · NATS JetStream

- Built a self-hostable event-driven workflow automation engine with parallel workers handling 1,000+ jobs/hour across multi-step automation pipelines.
- Enforced retries, idempotency, and failure isolation; used NATS JetStream to guarantee at-least-once delivery semantics.

### HTTP Server from Scratch [GitHub](#) Golang · TCP Sockets · HTTP/1.1

- Engineered a fully functional HTTP/1.1 server from raw TCP sockets; handled concurrent connections with goroutines and complete connection lifecycle control.
- Benchmarked at 3,000 requests/second locally, validating low-level networking and OS fundamentals.

### PRISM – University Platform Backend [GitHub](#) TypeScript · Node.js · PostgreSQL · Prisma · Docker · Nginx

- Built backend services for a university-wide platform with 1,000+ users; designed JWT authentication and RBAC with clean, testable service boundaries.
- Deployed containerized services behind Nginx with SSL termination, achieving 99%+ uptime across production.

## EDUCATION

### B.Tech in Computer Science Engineering | CGPA: 8.0 / 10.0

University of Lucknow, Lucknow, India

2022 – 2026

(Completed)

## ACHIEVEMENTS

- Maintained 8.0 CGPA throughout B.Tech CSE while independently shipping 3+ production-grade backend systems and contributing to two active open source projects simultaneously.